Skills Summary

One Solution, No Solution, or Infinitely Many

Use this reference while completing your worksheet or when studying.

The method never changes: **distribute, combine like terms, then collect the variable on one side**. What is left when the variable terms cancel (or fail to cancel) tells you which of the three cases you are in.

Skill 1 — One solution: the variable survives

Sign: after simplifying, the two sides have *different* x-terms (for example $5x$ against $3x$).

Subtracting the smaller x-term leaves a real equation to finish, and exactly one value of x makes it true. In a context, the two quantities are equal at exactly one moment.

Example: Classify and solve $5x + 2 = 3x + 10$.
Step 1. The x-terms differ ($5x$ against $3x$), so the variable cannot cancel — expect exactly one solution.
Step 2. Subtract $3x$: $2x + 2 = 10$. Subtract 2: $2x = 8$. Divide by 2: $\boldsymbol{x = 4}$ — one solution.

Try it: Classify and solve $7x - 4 = 2x + 11$.

check: one solution, $\boldsymbol{x = 3}$

Skill 2 — No solution: a false statement is left

Sign: after simplifying, the x-terms are *identical* but the constants are *different*.

The variable cancels completely and leaves something false, such as $12 = 7$. No value of x can fix a false statement, so the equation has no solution. In a context, two quantities that change at the same rate but start apart never meet.

Example: Classify $3(x + 4) = 3x + 7$.
Step 1. Distribute first — only then can the x-terms on the two sides be compared.
Step 2. $3x + 12 = 3x + 7$. Subtract $3x$: $12 = 7$, which is false, so no solution.

Try it: Classify $5x - 2 = 5x + 6$.

check: no solution

Skill 3 — Infinitely many: a true statement is left

Sign: after simplifying, the two sides are the *same expression* — same x-term *and* same constant.

The variable cancels and leaves something true, such as $0 = 0$. Every value of x works. In a context, the two expressions were two ways of writing the same quantity.

Example: Classify $4(2x - 3) = 8x - 12$.
Step 1. The right side looks like the expanded left side, so expand and compare rather than solving.
Step 2. $4(2x - 3) = 8x - 12$, so the equation reads $\mathbf{8x - 12 = 8x - 12}$ — true for every x, infinitely many solutions.

Try it: Classify $-3(x - 5) = 15 - 3x$.

check: $\mathbf{15 - 3x = 15 - 3x}$, infinitely many

Skill 4 — Building an equation to order

Working backwards: to force a case, control the two sides.

- **No solution:** make the x-terms match, the constants differ.
- **Infinitely many:** make the x-terms *and* the constants match.
- **One solution:** make the x-terms differ (the constants do not matter).

Example: Find a so that $5(2x - 3) = ax + 1$ has no solution.
Step 1. No solution means the x-terms must match, so expand the left side and read off its x-term.
Step 2. $5(2x - 3)$ expands to $\mathbf{10x - 15,}$ so $\boldsymbol{a = 10}$; the constants -15 and 1 differ, so the leftover statement is false.

Try it: Find a so that $4(3x + 2) = ax + 5$ has no solution.

check: $\mathbf{12x + 8,}$ so $\boldsymbol{a = 12}$